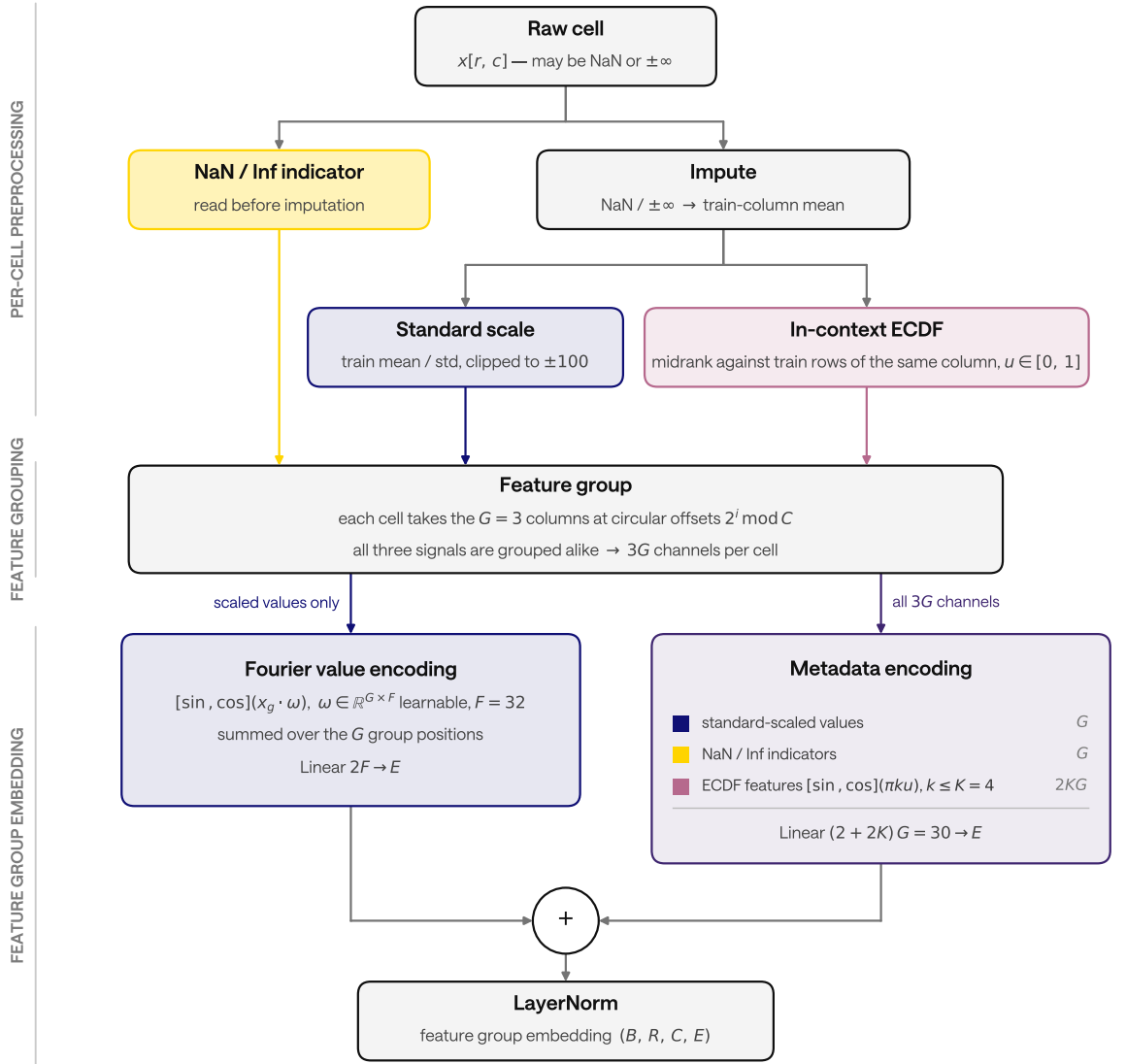

PER-CELL PREPROCESSING
FEATURE GROUPING
FEATURE GROUP EMBEDDING
Raw cell
x[r, c] — may be NaN or ±∞
NaN / Inf indicator
read before imputation
Impute
NaN / ±∞ → train-column mean
Standard scale
train mean / std, clipped to ±100
In-context ECDF
midrank against train rows of the same column, u ∈ [0, 1]
Feature group
each cell takes the G = 3 columns at circular offsets 2^i mod C
all three signals are grouped alike → 3G channels per cell
scaled values only
all 3G channels
Fourier value encoding
[sin , cos](x_g · ω), ω ∈ ℝ^{G × F} learnable, F = 32
summed over the G group positions
Linear 2F → E
Metadata encoding
standard-scaled values G
NaN / Inf indicators G
ECDF features [sin , cos](πku), k ≤ K = 4 2KG
Linear (2 + 2K) G = 30 → E
+
LayerNorm
feature group embedding (B, R, C, E)